

DHIR AGRAWAL

✉ dhiragrawal05@gmail.com  Dhir Agrawal  dhir-1

Projects

Lexis — Multimodal ASL & Speech Translation Platform

Key Skills: PyTorch, RTMPose, OpenCV, Groq API, FastAPI, PostgreSQL

Jun 2026 – Present

- Built a real-time ASL vision engine using RTMPose Wholebody pose estimation (133 keypoints, 30+ FPS) with a custom kinematic state machine for gesture segmentation and full A–Z fingerspelling fallback.
- Trained a PyTorch bidirectional 2-layer GRU on 2,400 ASL sign classes (WLASL + ASL Citizen), reaching 52% Top-1 and 74%+ Top-5 accuracy.
- Developed a multilingual speech-to-English translation pipeline (Whisper Large V3 Turbo + LLM translation) with sub-120ms transcription latency across 20+ spoken languages.
- Built a custom OpenCV-based data collection tool paired with an ExtraTrees ensemble classifier, expanding small user-recorded samples 25x through temporal augmentation.

The Corpus — Multi-Agent Financial RAG Research Project

Key Skills: Python, LangGraph, Groq, ChromaDB, FlashRank, rank_bm25, FastAPI

 GitHub Repository

Aug 2026 – Present

- Researched whether multi-agent RAG architectures justify their added complexity versus naive RAG baselines, across 10 companies' FY2024 10-K filings.
- Re-architected a 5-node LangGraph pipeline into a deterministic single-pass retrieval system, cutting query latency 9x (45–70s → 5.6–7.0s) with no loss of accuracy.
- Solved cross-company "single-document hijacking" via balanced entity-dispatch retrieval, and added deterministic post-generation numeric/date verification to gate against hallucinated figures.
- Identified and corrected multiple evaluation bugs (mismatched chunk budgets, unequal citation prompting) to produce a rigorous, reproducible naive-vs-multi-agent benchmark.

Dhir's Pit Wall — F1 2026 Race Prediction Dashboard

Key Skills: Python, XGBoost, FastF1, FastAPI, React, Vite, Tailwind CSS

 GitHub Repository  Live Demo

May 2026 – Present

- Built with a collaborator; engineered 13 pre-race features per driver from live F1 telemetry via the FastF1 API, including grid position, rolling form, and qualifying gap.
- Trained an XGBoost classifier with Leave-One-Race-Out cross-validation, achieving a 0.857 F1 score and 0.968 AUC on 2026 season data.
- Shipped a FastAPI backend with in-memory caching and a React dashboard exposing live predictions, feature importances, and race-by-race validation against actual results.

Education

Veer Narmad South Gujarat University

Bachelor of Computer Applications (BCA); CGPA: 7.83, Final Sem. SGPA: 8.82

Jun 2023 – May 2026

Surat, Gujarat

Achievements & Certificates

- ROGII Wellbore Geology Prediction — Kaggle competition on gamma-ray well-log TVT prediction; bronze medal, ranked 342nd of 6,125, using particle filtering, beam search, and LightGBM
- DeepLearning.AI Machine Learning Specialization — Andrew Ng, Stanford Online / DeepLearning.AI (Supervised Learning, Advanced Learning Algorithms, Unsupervised Learning & Recommender Systems)
- Anthropic AI Fluency Certificate

Languages & Technologies

Programming Languages: Python, JavaScript, SQL, C

ML / Deep Learning: PyTorch, scikit-learn, XGBoost, LightGBM, pandas, NumPy

LLM / RAG: LangGraph, ChromaDB, spaCy, rank_bm25, FlashRank

Computer Vision: RTMPose, OpenCV, MediaPipe

Backend: FastAPI, REST APIs, WebSockets

Frontend: React, Vite, Tailwind CSS

Database: PostgreSQL, MySQL, Firebase

Tools: Git, GitHub, Google Colab, Postman